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# Iris Flower Classification using Naive Bayes
# Google Colab Compatible
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# Step 1: Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Step 2: Load Iris Dataset
iris = load_iris()

# Create DataFrame
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['Species'] = iris.target

print("First Five Records")
print(df.head())

# Step 3: Define Features and Target
X = iris.data
y = iris.target

# Step 4: Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Step 5: Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Step 6: Train Naive Bayes Model
model = GaussianNB()
model.fit(X_train, y_train)

# Step 7: Prediction
y_pred = model.predict(X_test)

# Step 8: Evaluate Model
print("\nAccuracy:", accuracy_score(y_test, y_pred))

print("\nClassification Report")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix")
print(confusion_matrix(y_test, y_pred))

# Step 9: Predict a New Flower
new_flower = [[5.1, 3.5, 1.4, 0.2]]

new_flower_scaled = scaler.transform(new_flower)

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prediction = model.predict(new_flower_scaled)

print("\nPredicted Flower Species:",
      iris.target_names[prediction][0])

# Step 10: Visualization
plt.figure(figsize=(7,5))
plt.scatter(df.iloc[:,0], df.iloc[:,2],
           c=iris.target, cmap='viridis')

plt.xlabel("Sepal Length (cm)")
plt.ylabel("Petal Length (cm)")
plt.title("Iris Flower Classification")
plt.colorbar(label="Species")
plt.show()
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First Five Records

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

Species

0	0
1	0
2	0
3	0
4	0

Accuracy: 1.0

Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Confusion Matrix

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[[10 0 0]
 [ 0 9 0]
 [ 0 0 11]]
```

Predicted Flower Species: setosa

Iris Flower Classification

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2.00